Contents

1 Vector Spaces

When we read or hear the word *vector*, we may immediately think of two points in $\mathbb{R}^2$ (or $\mathbb{R}^3$) connected by an arrow. Mathematically speaking, a vector is just an element of a vector space. This then begs the question: *What is a vector space?*

Roughly speaking, a vector space is a set of objects that can be added and multiplied by scalars. We may have already worked with several types of vector spaces. Examples of vector spaces that we might have already encountered are:

1. the set $\mathbb{R}^n$,
2. the set of all $n \times n$ matrices,
3. the set of all functions from $[a, b]$ to $\mathbb{R}$, and
4. the set of all sequences.

In all of these sets, there is an operation of *addition* and an operation *multiplication by scalars*. Let us formalize exactly what we mean by a *vector space*.

Definition 1. *A nonempty set V (called a set of vectors) with operations "**vector addition**", denoted by "$+$" and "**scalar multiplication**", denoted by "$\cdot$" is called a **vector space** over a set of scalars $\mathbb{F}$ if the following properties are satisfied:*

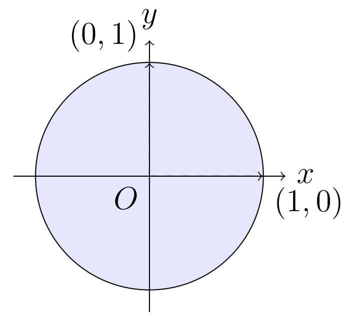
1. $u + v \in V$ *for all* $u, v \in V$. *(closure under addition)*
2. $(u + v) + w = u + (v + w)$. *(vector addition is associative)*
3. *There is a vector in V called the **zero vector**, denoted by $\mathbf{0}$, satisfying* $v + \mathbf{0} = v = \mathbf{0} + v$.
4. *For each v, there is a vector V, denoted by $-v$ such that* $v + (-v) = \mathbf{0} = (-v) + v$.
5. $u + v = v + u$ *for all* $u, v \in V$ *(addition is commutative)*
6. $\alpha \cdot v \in V$ *for every* $\alpha \in \mathbb{F}$ *and* $v \in V$. *(closure under scalar multiplication)*
7. $\alpha \cdot (\beta \cdot v) = (\alpha\beta) \cdot v$ *for every* $\alpha, \beta \in \mathbb{F}$ *and* $v \in V$.
8. $\alpha \cdot (u + v) = \alpha \cdot u + \alpha \cdot v$ *for every* $\alpha \in \mathbb{F}$ *and* $u, v \in V$.
9. $(\alpha + \beta) \cdot v = \alpha \cdot v + \beta \cdot v$ *for every* $\alpha, \beta \in \mathbb{F}$ *and* $v \in V$..
10. *There is an element* $1 \in \mathbb{F}$ *such that* $1 \cdot v = v$.

We usually take $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$. It can be shown that $0 \cdot \mathbf{v} = \mathbf{0}$ for any vector $\mathbf{v}$ in V, where $0 \in \mathbb{F}$. To better understand the definition of a vector space, we first consider a few elementary examples.

Example 2. *Let V be the unit disc in $\mathbb{R}^2$:*

$$V = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}.$$

Is V a vector space over $\mathbb{R}$?

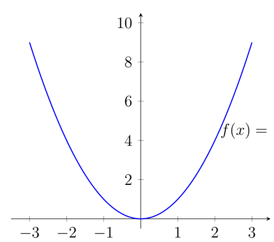


Solution. *The circle is not closed under scalar multiplication. For example, take $u = (1,0) \in V$ and multiply by say $\alpha = 2$. Then $\alpha u = (2,0)$ is not in V. Therefore, property (6) of the definition of a vector space fails, and consequently the unit disc is not a vector space.*

Example 3. *Let V be the graph of the quadratic function $f(x) = x^2$:*

$$V = \{(x,y) \in \mathbb{R}^2 \mid y = x^2\}.$$

Is V a vector space over $\mathbb{R}$?

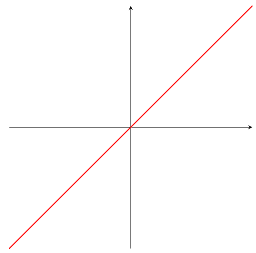


Solution. *The set V is not closed under scalar multiplication. For example, $\mathbf{u} = (1,1)$ is a point in V but $2\mathbf{u} = (2,2)$ is not. You may also notice that V is not closed under addition either. For example, both $\mathbf{u} = (1,1)$ and $\mathbf{v} = (2,4)$ are in V but $\mathbf{u} + \mathbf{v} = (3,5)$ and $(3,5)$ is not a point on the parabola V. Therefore, the graph of $f(x) = x^2$ is not a vector space.*

Example 4. *Let V be the graph of the function $f(x) = 2x$:*

$$V = \{(x,y) \in \mathbb{R}^2 \mid y = 2x\}.$$

Is V a vector space over $\mathbb{R}$?



Solution. *We will show that V is a vector space. First, we verify that V is closed under addition. We first note that an arbitrary point in V can be written as $\mathbf{u} = (x, 2x)$. Let then $\mathbf{u} = (a, 2a)$ and $\mathbf{v} = (b, 2b)$ be points in V. Then*

$$\mathbf{u} + \mathbf{v} = (a + b, 2a + 2b) = (a + b, 2(a + b)).$$

Therefore, V is closed under addition. Verify that V is closed under scalar multiplication:

$$\alpha\mathbf{u} = \alpha(a, 2a) = (\alpha a, \alpha 2a) = (\alpha a, 2(\alpha a)).$$

Therefore, V is closed under scalar multiplication. There is a zero vector $\mathbf{0} = (0, 0)$ in V:

$$\mathbf{u} + \mathbf{0} = (a, 2a) + (0, 0) = (a, 2a).$$

All the other properties of a vector space can be verified to hold; for example, addition is commutative and associative in V because addition in $\mathbb{R}^2$ is commutative/associative, etc. Therefore, the graph of the function $f(x) = 2x$ is a vector space.

Example 5. *Let $V = \mathbb{R}_n[t]$ be the set of all polynomials in the variable t of degree at most n, given by*

$$\mathbb{R}_n[t] = \left\{a_0 + a_1 t + a_2 t^2 + \cdots + a_n t^n \mid a_0, a_1, \ldots, a_n \in \mathbb{R}\right\}.$$

Is V a vector space over $\mathbb{R}$?

Solution. *Let $u(t) = u_0 + u_1 t + \cdots + u_n t^n$ and $v(t) = v_0 + v_1 t + \cdots + v_n t^n$ be polynomials in V. We define the addition of u and v as the new polynomial $(u + v)$ as follows:*

$$(u + v)(t) = u(t) + v(t) = (u_0 + v_0) + (u_1 + v_1)t + \cdots + (u_n + v_n)t^n.$$

Then $u + v$ is a polynomial of degree at most n, and thus $(u + v) \in \mathbb{R}_n[t]$. Therefore, $\mathbb{R}_n[t]$ is closed under addition.

Now let α be a scalar. Define a new polynomial (αu) as follows:

$$(\alpha u)(t) = (\alpha u_0) + (\alpha u_1)t + \cdots + (\alpha u_n)t^n.$$

Then αu is a polynomial of degree at most n, and thus $\alpha u \in \mathbb{R}_n[t]$. Hence, $\mathbb{R}_n[t]$ is closed under scalar multiplication. The zero vector in $\mathbb{R}_n[t]$ is the zero polynomial $0(t) = 0$.

One can verify that all other properties of the definition of a vector space also hold; for example, addition is commutative and associative, etc. Thus, $\mathbb{R}_n[t]$ is a vector space over $\mathbb{R}$.

Example 6. *Let $V = M_{m\times n}(\mathbb{R})$ be the set of all $m \times n$ matrices whose entries are from $\mathbb{R}$. Under the usual operations of addition of matrices and scalar multiplication, is $M_{m\times n}$ a vector space over $\mathbb{R}$?*

Solution. Given matrices $A, B \in M_{m\times n}(\mathbb{R})$ and a scalar α, we define the sum $A + B$ by adding entry-by-entry, and αA by multiplying each entry of A by α. It is clear that the space $M_{m\times n}(\mathbb{R})$ is closed under these two operations. The zero vector in $M_{m\times n}(\mathbb{R})$ is the matrix of size $m \times n$ having all entries equal to zero. It can be verified that all other properties of the definition of a vector space also hold. Thus, the set $M_{m\times n}(\mathbb{R})$ is a vector space over $\mathbb{R}$.

Example 7. *The n-dimensional Euclidean space $V = \mathbb{R}^n$ under the usual operations of addition and scalar multiplication is a vector space over $\mathbb{R}$.*

Example 8. *Let $V = C[a, b]$ denote the set of functions with domain $[a, b]$ and codomain $\mathbb{R}$ that are continuous. Is V a vector space over $\mathbb{R}$?*

1.1 Subspaces of vector spaces

Frequently, one encounters a vector space W that is a subset of a larger vector space V. In this case, we would say that W is a subspace of V. Below is the formal definition.

Definition 9. *Let V be a vector space over $\mathbb{F}$ with operations "$+$" and "$\cdot$" and let $W \subseteq V$. Then W is called a subspace of V if W is a vector space under same operations $+$ and $\cdot$ over $\mathbb{F}$.*

Since W is also a vector space, all 10-properties of the vector space hold. Hence it is closed under vector addition and scalar multiplication. The following theorem says that without checking all the 10-properties to show a subset W of V is a subspace of V, we can only check 2-properties.

Theorem 10. *Let V be a vector space over $\mathbb{F}$ under operations "$+$" and "$\cdot$". A subset W of V is a **subspace** of V if and only if it satisfies the following properties:*

1. $u + v \in W$ *for all* $u, v \in W$
2. $\alpha \cdot u \in W$ *for all* $\alpha \in \mathbb{F}$ *and* $u \in W$.

Example 11. *Let W be the graph of the function $f(x) = 2x$:*

$$W = \{(x, y) \in \mathbb{R}^2 \mid y = 2x\}.$$

Is W a subspace of $V = \mathbb{R}^2$?

Solution. If $x = 0$, then $y = 2 \cdot 0 = 0$ and therefore $(0, 0)$ is in W. Let $\mathbf{u} = (a, 2a)$ and $\mathbf{v} = (b, 2b)$ be elements of W. Then

$$\mathbf{u} + \mathbf{v} = (a, 2a) + (b, 2b) = (a + b, 2a + 2b) = (a + b, 2(a + b)).$$

Because the x- and y-components of $\mathbf{u} + \mathbf{v}$ satisfy $y = 2x$, then $\mathbf{u} + \mathbf{v}$ is inside W. Thus, W is closed under addition.

Let α be any scalar and let $\mathbf{u} = (a, 2a)$ be an element of W. Then

$$\alpha\mathbf{u} = (\alpha a, \alpha 2a) = (\alpha a, 2(\alpha a)).$$

Because the x- and y-components of $\alpha\mathbf{u}$ satisfy $y = 2x$, then $\alpha\mathbf{u}$ is an element of W, and thus W is closed under scalar multiplication.

All three conditions of a subspace are satisfied for W, and therefore W is a subspace of V.

Example 12. *Let W be the first quadrant in $\mathbb{R}^2$:*

$$W = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0, y \geq 0\}.$$

Is W a subspace of $\mathbb{R}^2$?

Solution. The set W contains the zero vector, and the sum of two vectors in W is again in W; you may want to verify this explicitly as follows: If $\mathbf{u}_1 = (x_1, y_1)$ is in W, then $x_1 \geq 0$ and $y_1 \geq 0$, and *similarly if* $\mathbf{u}_2 = (x_2, y_2)$ *is in* W*, then* $x_2 \geq 0$ *and* $y_2 \geq 0$*. Then the sum*

$$\mathbf{u}_1 + \mathbf{u}_2 = (x_1 + x_2, y_1 + y_2)$$

has components $x_1 + x_2 \geq 0$ *and* $y_1 + y_2 \geq 0$*, and therefore* $\mathbf{u}_1 + \mathbf{u}_2$ *is in* W*. However,* W *is not closed under scalar multiplication. For example, if* $\mathbf{u} = (1, 1)$ *and* $\alpha = -1$*, then*

$$\alpha\mathbf{u} = (-1, -1)$$

is not in W *because the components of* $\alpha\mathbf{u}$ *are clearly not non-negative.*

Example 13. *Let* $V = M_{n\times n}$ *be the vector space of all* $n \times n$ *matrices. We define the trace of a matrix* $A \in M_{n\times n}$ *as the sum of its diagonal entries:*

$$\mathit{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}.$$

Let W *be the set of all* $n \times n$ *matrices whose trace is zero:*

$$W = \{A \in M_{n\times n} \mid \mathit{tr}(A) = 0\}.$$

Is W *a subspace of* V*?*

Solution. If 0 *is the* $n \times n$ *zero matrix, then clearly* $\mathit{tr}(0) = 0$*, and thus* $0 \in M_{n\times n}$*. Suppose that* A *and* B *are in* W*. Then necessarily* $\mathit{tr}(A) = 0$ *and* $\mathit{tr}(B) = 0$*. Consider the matrix*

$$C = A + B.$$

Then

$$\mathit{tr}(C) = \mathit{tr}(A + B) = (a_{11} + b_{11}) + (a_{22} + b_{22}) + \cdots + (a_{nn} + b_{nn})$$

$$= (a_{11} + \cdots + a_{nn}) + (b_{11} + \cdots + b_{nn}) = \mathit{tr}(A) + \mathit{tr}(B) = 0$$

Therefore, $\mathit{tr}(C) = 0$ *and consequently* $C = A + B \in W$*, in other words,* W *is closed under addition. Now let* α *be a scalar and let* $C = \alpha A$*. Then*

$$\mathit{tr}(C) = \mathit{tr}(\alpha A) = (\alpha a_{11}) + (\alpha a_{22}) + \cdots + (\alpha a_{nn}) = \alpha\, \mathit{tr}(A) = 0.$$

Thus, $\mathit{tr}(C) = 0$*, that is,* $C = \alpha A \in W$*, and consequently* W *is closed under scalar multiplication. Therefore, the set* W *is a subspace of* V*.*

Example 14. *Let* $V = P_n[t]$ *and consider the subset* W *of* V*:*

$$W = \{u \in P_n[t] \mid u(2) = -1\}$$

In other words, W *consists of polynomials of degree* n *in the variable* t *whose value at* $t = 2$ *is* -1*. Is* W *a subspace of* V*?*

Solution. The zero polynomial $0(t) = 0$ *clearly does not equal* -1 *at* $t = 2$*. Therefore,* W *does not contain the zero polynomial and, because all three conditions of a subspace must be satisfied for* W *to be a subspace, we conclude that* W *is not a subspace of* $n[t]$*. As an exercise, you may want to investigate whether or not* W *is closed under addition and scalar multiplication.*

Example 15. *A square matrix* A *is said to be symmetric if* $A^T = A$*. For example, the following is a* 3×3 *symmetric matrix:*

$$A = \begin{pmatrix} 2 & 4 & 5 \\ 1 & 2 & -3 \\ -3 & 5 & 7 \end{pmatrix}$$

Verify for yourself that we do indeed have that $A^T = A$*. Let* W *be the set of all symmetric* $n \times n$ *matrices. Is* W *a subspace of* $V = M_{n \times n}$*?*

Example 16. *For any vector space* V*, there are two trivial subspaces in* V*, namely,* V *itself is a subspace of* V*, and the set consisting of the zero vector,* $W = \{0\}$*, is a subspace of* V*.*

There is one particular way to generate a subspace of any given vector space V using the span of a set of vectors. Recall that we defined the span of a set of vectors in n, but we can define the same notion on a general vector space V.

Definition 17. *Let* V *be a vector space and let* $v_1, v_2, \ldots, v_p$ *be vectors in* V*. The span of* $\{v_1, v_2, \ldots, v_p\}$ *is the set of all linear combinations of* $v_1, v_2, \ldots, v_p$*:*

$$\text{span}\{v_1, v_2, \ldots, v_p\} = \{t_1 v_1 + t_2 v_2 + \cdots + t_p v_p \mid t_1, t_2, \ldots, t_p \in \mathbb{R}\}.$$

We now show that the span of a set of vectors in V is a subspace of V.

Theorem 18. *If* $v_1, v_2, \ldots, v_p$ *are vectors in* V*, then span*$\{v_1, v_2, \ldots, v_p\}$ *is a subspace of* V*.*

Proof. Let $u = t_1v_1 + \cdots + t_pv_p$ and $w = s_1v_1 + \cdots + s_pv_p$ be two vectors in span$\{v_1, v_2, \ldots, v_p\}$. Then

$$u + w = (t_1v_1 + \cdots + t_pv_p) + (s_1v_1 + \cdots + s_pv_p) = (t_1 + s_1)v_1 + \cdots + (t_p + s_p)v_p.$$

Therefore, $u + w$ is also in the span of $v_1, \ldots, v_p$. Now consider αu:

$$\alpha u = \alpha(t_1v_1 + \cdots + t_pv_p) = (\alpha t_1)v_1 + \cdots + (\alpha t_p)v_p.$$

Therefore, αu is in the span of $v_1, \ldots, v_p$. Lastly, since $0v_1 + 0v_2 + \cdots + 0v_p = 0$, the zero vector 0 is in the span of $v_1, v_2, \ldots, v_p$. Therefore, $span\{v_1, v_2, \ldots, v_p\}$ is a subspace of V.

Given a general subspace W of V, if $w_1, w_2, \ldots, w_p$ are vectors in W such that

$$\text{span}\{w_1, w_2, \ldots, w_p\} = W,$$

then we say that $\{w_1, w_2, \ldots, w_p\}$ is a spanning set of W. Hence, every vector in W can be written as a linear combination of the vectors $w_1, w_2, \ldots, w_p$. □

1.2 Linear independence

Roughly speaking, the concept of linear independence evolves around the idea of working with "efficient" spanning sets for a subspace. For instance, the set of directions $\{EAST, NORTH, NORTH-EAST\}$ are redundant since a total displacement in the NORTH-EAST direction can be obtained by combining individual NORTH and EAST displacements. With these vague statements out of the way, we introduce the formal definition of what it means for a set of vectors to be "efficient".

Definition 19. *Let V be a vector space over $\mathbb{F}$ and let $\{v_1, v_2, \ldots, v_p\}$ be a set of vectors in V. Then $\{v_1, v_2, \ldots, v_p\}$ is linearly independent if the only scalars $c_1, c_2, \ldots, c_p \in \mathbb{F}$ that satisfy the equation*

$$c_1v_1 + c_2v_2 + \cdots + c_pv_p = 0$$

are the trivial scalars $c_1 = c_2 = \cdots = c_p = 0$. If the set $\{v_1, \ldots, v_p\}$ is not linearly independent, then we say that it is linearly dependent.

An infinite set S of vectors in V is said to be linearly independent if every finite subset of S is linearly independent.

We now describe the redundancy in a set of linearly dependent vectors. If $\{v_1, \ldots, v_p\}$ are linearly dependent, it follows that there are scalars $c_1, c_2, \ldots, c_p$, at least one of which is nonzero, such that

$$c_1 v_1 + c_2 v_2 + \cdots + c_p v_p = 0. \tag{$\star$}$$

For example, suppose that $\{v_1, v_2, v_3, v_4\}$ are linearly dependent. Then there are scalars c_1, c_2, c_3, c_4, not all of them zero, such that equation $(\star)$ holds. Suppose, for the sake of argument, that $c_3 \neq 0$. Then,

$$v_3 = -\frac{c_1}{c_3} v_1 - \frac{c_2}{c_3} v_2 - \frac{c_4}{c_3} v_4.$$

Therefore, when a set of vectors is linearly dependent, it is possible to write one of the vectors as a linear combination of the others. It is in this sense that a set of linearly dependent vectors is redundant. In fact, if a set of vectors is linearly dependent, we can say even more, as the following theorem states.

Theorem 20. *A set of vectors $\{v_1, v_2, \ldots, v_p\}$, with $v_1 \neq 0$, is linearly dependent if and only if some v_j is a linear combination of the preceding vectors $v_1, \ldots, v_{j-1}$.*

Example 21. *Show that the following set of 2×2 matrices is linearly dependent:*

$$A_1 = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} -1 & 3 \\ 1 & 0 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 5 & 0 \\ -2 & -3 \end{bmatrix}.$$

Solution: It is clear that A_1 and A_2 are linearly independent, i.e., A_1 cannot be written as a scalar multiple of A_2, and vice versa. Since the $(2,1)$ entry of A_1 is zero, the only way to get the -2 in the $(2,1)$ entry of A_3 is to multiply A_2 by -2. Similarly, since the $(2,2)$ entry of A_2 is zero, the only way to get the -3 in the $(2,2)$ entry of A_3 is to multiply A_1 by 3. Hence, we suspect that $3A_1 - 2A_2 = A_3$. Verify:

$$3A_1 - 2A_2 = \begin{bmatrix} 3 & 6 \\ 0 & -3 \end{bmatrix} - \begin{bmatrix} -2 & 6 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ -2 & -3 \end{bmatrix} = A_3.$$

Therefore, $3A_1 - 2A_2 - A_3 = 0$, and thus we have found scalars c_1, c_2, c_3, not all zero, such that $c_1 A_1 + c_2 A_2 + c_3 A_3 = 0$.

1.3 Basis of a vector space

We now introduce the important concept of a basis. Given a set of vectors $\{v_1, \ldots, v_{p-1}, v_p\}$ in V, we showed that $W = \mathrm{span}\{v_1, v_2, \ldots, v_p\}$ is a subspace of V. If, say, v_p is linearly dependent on $v_1, v_2, \ldots, v_{p-1}$, then we can remove v_p and the smaller set $\{v_1, \ldots, v_{p-1}\}$ still spans all of W:

$$W = \text{span}\{v_1, v_2, \ldots, v_p\} = \text{span}\{v_1, \ldots, v_{p-1}\}.$$

Intuitively, v_p does not provide an independent "direction" in generating W. If some other vector v_j is linearly dependent on $v_1, \ldots, v_{p-1}$, then we can remove v_j and the resulting smaller set of vectors still spans W. We can continue removing vectors until we obtain a minimal set of vectors that are linearly independent and still span W. The following remarks motivate the following important definition.

Definition 22. *Let W be a subspace of a vector space V. A set of vectors $\mathcal{B} = \{v_1, \ldots, v_k\}$ in W is said to be a basis for W if:*

1. *The set $\mathcal{B}$ spans all of W, that is, $W = \mathrm{span}\{v_1, \ldots, v_k\}$, and*

2. *The set $\mathcal{B}$ is linearly independent.*

A basis is therefore a *minimal spanning set* for a subspace. Indeed, if $\mathcal{B} = \{v_1, \ldots, v_p\}$ is a basis for W and we remove, say, v_p, then

$$\widetilde{\mathcal{B}} = \{v_1, \ldots, v_{p-1}\}$$

cannot be a basis for W (*Why?*). If $\mathcal{B} = \{v_1, \ldots, v_p\}$ is a basis, then it is linearly independent, and therefore v_p cannot be written as a linear combination of the others. In other words, $v_p \in W$ is not in the span of

$$\widetilde{\mathcal{B}} = \{v_1, \ldots, v_{p-1}\},$$

and therefore $\widetilde{\mathcal{B}}$ is not a basis for W, because a basis must be a spanning set.

If, on the other hand, we start with a basis $\mathcal{B} = \{v_1, \ldots, v_p\}$ for W and we add a new vector u from W, then

$$\widetilde{\mathcal{B}} = \{v_1, \ldots, v_p, u\}$$

is not a basis for W (*Why?*). We still have that $\text{span}(\widetilde{\mathcal{B}}) = W$, but now $\widetilde{\mathcal{B}}$ is not linearly independent. Indeed, because $\mathcal{B} = \{v_1, \ldots, v_p\}$ is a basis for W, the vector u can be written as a linear combination of $\{v_1, \ldots, v_p\}$, and thus $\widetilde{\mathcal{B}}$ is not linearly independent.

Example 23. *Show that the standard unit vectors form a basis for $V = \mathbb{R}^3$:*

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Solution: Any vector $x \in \mathbb{R}^3$ can be written as a linear combination of e_1, e_2, e_3:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = x_1 e_1 + x_2 e_2 + x_3 e_3.$$

Therefore, span$\{e_1, e_2, e_3\} = \mathbb{R}^3$. The set $\mathcal{B} = \{e_1, e_2, e_3\}$ is linearly independent. Indeed, if there are scalars c_1, c_2, c_3 such that

$$c_1 e_1 + c_2 e_2 + c_3 e_3 = 0,$$

then clearly they must all be zero: $c_1 = c_2 = c_3 = 0$. Therefore, by definition, $\mathcal{B} = \{e_1, e_2, e_3\}$ is a basis for $\mathbb{R}^3$. This basis is called the standard basis for $\mathbb{R}^3$. Analogous arguments hold for $\{e_1, e_2, \ldots, e_n\}$ in $\mathbb{R}^n$.

Example 24. *Is $B = \{v_1, v_2, v_3\}$ a basis for $\mathbb{R}^3$?*

$$v_1 = \begin{bmatrix} 2 \\ -2 \\ -4 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -4 \\ 8 \\ 0 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 4 \\ -6 \\ -6 \end{bmatrix}.$$

Solution: Form the matrix $A = [v_1\ v_2\ v_3]$ and row reduce:

$$A \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore, the only solution to $Ax = 0$ is the trivial solution. Thus, $\mathcal{B}$ is linearly independent. Moreover, for any $b \in \mathbb{R}^3$, the augmented matrix $[A\ b]$ is consistent. Therefore, the columns of A

span all of $\mathbb{R}^3$*:*

$$Col(A) = span\{v_1, v_2, v_3\} = \mathbb{R}^3.$$

Hence, $\mathcal{B}$ *is a basis for* $\mathbb{R}^3$.

Example 25. *In* $V = \mathbb{R}^4$*, consider the vectors*

$$v_1 = \begin{bmatrix} 1 \\ 3 \\ 2 \\ -1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ -2 \\ -2 \\ 1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} -1 \\ 4 \\ 2 \\ -3 \end{bmatrix}.$$

Let $W = span\{v_1, v_2, v_3\}$*. Is* $\mathcal{B} = \{v_1, v_2, v_3\}$ *a basis for* W*?*

Solution: By definition, $\mathcal{B}$ *is a spanning set for* W*, so we need only determine if* $\mathcal{B}$ *is linearly independent. Form the matrix* $A = [v_1\ v_2\ v_3]$ *and row reduce to obtain:*

$$A \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Hence, $rank(A) = 2$ *and thus* $\mathcal{B}$ *is linearly dependent. Notice* $v_1 - v_2 = v_3$*. Therefore,* $\mathcal{B}$ *is not a basis of* W*.*

Example 26. *Find a basis for the vector space of* 2×2 *matrices.*

Example 27. *Recall that an* $n \times n$ *matrix* A *is skew-symmetric if* $A^\top = -A$*. We proved that the set of* $n \times n$ *skew-symmetric matrices is a subspace. Find a basis for the set of* 3×3 *skew-symmetric matrices.*

1.4 Dimension of a vector space

The following theorem will lead to the definition of the dimension of a vector space.

Theorem 28. *Any two bases of a vector space have equal cardinality.*

Proof. We will prove the theorem for the case that $V = \mathbb{R}^n$. We already know that the standard unit vectors $\{e_1, e_2, \ldots, e_n\}$ form a basis of $\mathbb{R}^n$. Let $\{u_1, u_2, \ldots, u_p\}$ be nonzero vectors in $\mathbb{R}^n$, and suppose first that $p > n$. In previous theorem, we proved that any set of vectors in $\mathbb{R}^n$ containing more than n vectors is automatically linearly dependent. The reason is that the RREF of $A = [u_1\, u_2\, \cdots\, u_p]$ will contain at most $r = n$ leading ones, and therefore $d = p - n > 0$. Thus, the solution set of $Ax = 0$ contains non-trivial solutions. On the other hand, suppose instead that $p < n$. Also, a set of vectors $\{u_1, \ldots, u_p\}$ in $\mathbb{R}^n$ spans $\mathbb{R}^n$ if and only if the RREF of A has exactly $r = n$ leading ones. The largest possible value of r is $r = p < n$. Therefore, if $p < n$, then $\{u_1, u_2, \ldots, u_p\}$ cannot be a basis for $\mathbb{R}^n$. Thus, in either case ($p > n$ or $p < n$), the set $\{u_1, u_2, \ldots, u_p\}$ cannot be a basis for $\mathbb{R}^n$. Hence, any basis in $\mathbb{R}^n$ must contain n vectors. $\square$

The previous theorem does not say that every set $\{v_1, v_2, \ldots, v_n\}$ of nonzero vectors in $\mathbb{R}^n$ containing n vectors is automatically a basis for $\mathbb{R}^n$. For example,

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 0 \\ 3 \\ 0 \\ 0 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

do not form a basis for $\mathbb{R}^3$ because

$$x = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

is not in the span of $\{v_1, v_2, v_3\}$. All that we can say is that a set of vectors in $\mathbb{R}^n$ containing fewer or more than n vectors is automatically not a basis for $\mathbb{R}^n$. From above Theorem 1, any basis in $\mathbb{R}^n$ must have exactly n vectors. In fact, on a general abstract vector space V, if $\{v_1, v_2, \ldots, v_n\}$ is a basis for V, then any other basis for V must have exactly n vectors also. Because of this result, we can make the following definition.

Definition 29. *Let V be a vector space. The dimension of V, denoted as $\dim V$, is the number of vectors in any basis of V. The dimension of the trivial vector space $V = \{0\}$ is defined to be zero.*

Moving on, suppose that we have a set $B = \{v_1, v_2, \ldots, v_n\}$ in $\mathbb{R}^n$ containing exactly n vectors. For $B = \{v_1, v_2, \ldots, v_n\}$ to be a basis of $\mathbb{R}^n$, the set B must be linearly independent and $\operatorname{span}(B) = \mathbb{R}^n$. In fact, it can be shown that if B is linearly independent, then the spanning condition $\text{span}(B) = \mathbb{R}^n$ is automatically satisfied, and vice-versa.

For example, say the vectors $\{v_1, v_2, \ldots, v_n\}$ in $\mathbb{R}^n$ are linearly independent, and put $A = [v_1\, v_2\, \cdots\, v_n]$. Then A^{-1} exists, and therefore $Ax = b$ is always solvable. Hence,

$$\text{Col}(A) = \text{span}\{v_1, v_2, \ldots, v_n\} = \mathbb{R}^n.$$

In summary, we have the following theorem.

Theorem 30. *Let $B = \{v_1, \ldots, v_n\}$ be vectors in $\mathbb{R}^n$. If B is linearly independent, then B is a basis for $\mathbb{R}^n$. Alternatively, if $\text{span}\{v_1, v_2, \ldots, v_n\} = \mathbb{R}^n$, then B is a basis for $\mathbb{R}^n$.*

Example 31. *Do the columns of the matrix*

$$A = \begin{bmatrix} 2 & 3 & 3 & -2 \\ 4 & 7 & 8 & -6 \\ 0 & 0 & 1 & 0 \\ -4 & -6 & -6 & 3 \end{bmatrix}$$

form a basis for $\mathbb{R}^4$?

Solution: Let v_1, v_2, v_3, v_4 denote the columns of A. Since we have $n = 4$ vectors in $\mathbb{R}^4$, we need only check that they are linearly independent. Compute

$$\det(A) = -2 \neq 0.$$

Hence, $\text{rank}(A) = 4$, and thus the columns of A are linearly independent. Therefore, the vectors v_1, v_2, v_3, v_4 form a basis for $\mathbb{R}^4$.

A subspace W of a vector space V is a vector space in its own right, and therefore also has a dimension. By definition, if $B = \{v_1, \ldots, v_k\}$ is a linearly independent set in W and $\text{span}\{v_1, \ldots, v_k\} = W$, then B is a basis for W, and in this case, the dimension of W is k. Since an n-dimensional vector space V requires exactly n vectors in any basis, then if W is a strict subspace of V, then

$$\dim(W) < \dim(V).$$

As an example, in $V = \mathbb{R}^3$, subspaces can be classified by dimension:

1. The zero-dimensional subspace in $\mathbb{R}^3$ is $W = \{0\}$.
2. The one-dimensional subspaces in $\mathbb{R}^3$ are lines through the origin. These are spanned by a single nonzero vector.
3. The two-dimensional subspaces in $\mathbb{R}^3$ are planes through the origin. These are spanned by two linearly independent vectors.
4. The only three-dimensional subspace in $\mathbb{R}^3$ is $\mathbb{R}^3$ itself. Any set $\{v_1, v_2, v_3\}$ in $\mathbb{R}^3$ that is linearly independent is a basis for $\mathbb{R}^3$.

Example 32. *Find a basis for* $\mathrm{Null}(A)$ *and the* $\dim(\mathrm{Null}(A))$ *if*

$$A = \begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix}.$$

Solution: By definition, $\mathrm{Null}(A)$ *is the solution set of the homogeneous system* $Ax = 0$. *Row reducing, we obtain*

$$A \sim \begin{bmatrix} 1 & 0 & 6 & 5 \\ 0 & 1 & 5/2 & 3/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The general solution to $Ax = 0$ *in parametric form is*

$$x = t \begin{bmatrix} -5 \\ -3/2 \\ 0 \\ 1 \end{bmatrix} + s \begin{bmatrix} -6 \\ -5/2 \\ 1 \\ 0 \end{bmatrix} = tv_1 + sv_2.$$

By construction, the vectors

$$v_1 = \begin{bmatrix} -5 \\ -3/2 \\ 0 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -6 \\ -5/2 \\ 1 \\ 0 \end{bmatrix}$$

span $\mathrm{Null}(A)$ *and are linearly independent. Therefore,* $B = \{v_1, v_2\}$ *is a basis for* $\mathrm{Null}(A)$*, and thus*

$$\dim(\mathrm{Null}(A)) = 2.$$

In general, the dimension of $\mathrm{Null}(A)$ *is the number of free parameters in the solution set of the system* $Ax = 0$*, that is,*

$$\dim(\mathrm{Null}(A)) = d = n - \mathrm{rank}(A).$$

1.5 Subspaces associated with a matrix

Given a matrix of size $m \times n$, we have some naturally associated vector spaces, which are discussed below.

1.5.1 Row space, column space and rank of a matrix

If A is an $m \times n$ matrix with entries in $\mathbb{F}$, each row of A has n entries and thus can be identified with a vector in $\mathbb{F}^n$. Similarly, each column of A has m entries, and can be identified with a vector in $\mathbb{F}^m$.

Definition 33.

- *The set of all linear combinations of the row vectors of* A *is a subspace of* $\mathbb{F}^n$*, called the row space of* A*. The dimension of the row space is said to be the row rank of* A*.*
- *The set of all linear combinations of the column vectors of* A *is a subspace of* $\mathbb{F}^m$*, called the row space of* A*. The dimension of the column space is said to be the column rank of* A*.*

Example 34. Let

$$A = \begin{bmatrix} -2 & -5 & 7 & 3 \\ 2 & 4 & -2 & 1 \\ 3 & 7 & -8 & 6 \end{bmatrix} \in M_{3\times 4}(\mathbb{R}), \quad \mathbf{b} = \begin{bmatrix} 3 \\ -1 \\ 3 \end{bmatrix}.$$

Is $\mathbf{b}$ in the column space of A?

The vector $\mathbf{b}$ is in the column space of A if there exists $\mathbf{x} \in \mathbb{R}^4$ such that $A\mathbf{x} = \mathbf{b}$. Hence, we must determine if $A\mathbf{x} = \mathbf{b}$ has a solution. Performing elementary row operations on the augmented matrix $[A \,|\, \mathbf{b}]$, we obtain:

$$[A \,|\, \mathbf{b}] \sim \begin{bmatrix} 0 & 1 & -5 & -4 & -2 \\ 2 & 4 & -2 & 1 & 3 \\ 0 & 0 & 0 & 17 & 1 \end{bmatrix}.$$

The system is consistent, and therefore $A\mathbf{x} = \mathbf{b}$ will have a solution. Thus, $\mathbf{b}$ is in the column space of A.

Recall that two matrices are row-equivalent when one can be obtained from the other by performing finitely many elementary row operations. One can see that row-equivalent matrices have the same row space.

We want to find a basis for the row space of A. If A is in row echelon form, then its non-zero row vectors are linearly independent, and hence form a basis for the row space of A. In general, we have the following useful result.

Theorem 35. *If a matrix A is row-equivalent to a matrix B in row echelon form, then the non-zero row vectors of B form a basis for the row space of A.*

Observe that, the column vectors of A are the row vectors of A^T. Hence, to a find a basis for the column space of A, we find a basis for the row space of A^T as above.

In fact, we have the following:

Theorem 36. *The row rank and column rank of any matrix is the same.*

Thus we may define the notion of rank of a matrix as follows:

Definition 37. *The dimension of the row (or column) space of a matrix is called the rank of A and is denoted by rank(A).*

1.5.2 Null space of a matrix

Let A be a matrix in $M_{m\times n}(\mathbb{F})$. Let us consider the homogeneous system of equations, $A\mathbf{x} = \mathbf{0}$, where $\mathbf{x} = [x_1, x_2, \ldots, x_n]^T$ be the column vector of unknowns. Then the set of all solutions of this homogeneous system of equations form a subspace of $\mathbb{F}^n$, called the null space of A, which we define below.

Definition 38. *The null space of a matrix $A \in M_{m\times n}(\mathbb{F})$, denoted by Null$(A)$, is the subset of $\mathbb{F}^n$ consisting of vectors $\mathbf{v} \in \mathbb{F}^n$ such that $A\mathbf{v} = \mathbf{0}$. Using set notation:*

$$\mathit{Null}(A) = \{\mathbf{v} \in \mathbb{F}^n \mid A\mathbf{v} = \mathbf{0}\}.$$

This is a subspace of $\mathbb{F}^n$. The dimension of the null space of A is called the nullity of A.

Example 39. Consider the following subset of $\mathbb{R}^4$:

$$W = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 \mid 2x_1 - 3x_2 + x_3 - 7x_4 = \mathbf{0}\}.$$

Is W a subspace of $\mathbb{R}^4$?

Yes! The set W is the null space of the 1×4 matrix A given by

$$A = \begin{bmatrix} 2 & -3 & 1 & -7 \end{bmatrix}.$$

Hence, $W = \text{Null}(A)$ and consequently W is a subspace.

Recall the linear transformation $L_A : \mathbb{F}^n \to \mathbb{F}^m$ of Example **??**. We see that, $\mathbf{v}$ is in the kernel of L_A if and only if $L_A(\mathbf{v}) = A\mathbf{v} = \mathbf{0}$. In other words, $\ker(L_A) = \text{Null}(A)$. Also, note that range of L_A is the column space of A. Hence, by the rank-nullity theorem applied to the linear transformation L_A, we have the following:

Theorem 40. *For a matrix $A \in M_{m\times n}(\mathbb{F})$, rank$(A)$ + nullity$(A) = n$.*

By the above result, if the the row reduced echelon form of A has r leading 1's, that is, rank of A is r, then dimension of the null space is $d = n - r$. Therefore, we will obtain vectors $\mathbf{v}_1, \ldots, \mathbf{v}_d$ such that

$$\text{Null}(A) = \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_d\}.$$

Example 41. Find a spanning set for the null space of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \in M_{3\times 5}(\mathbb{R}).$$

The null space of A is the solution set of the homogeneous system $A\mathbf{x} = \mathbf{0}$. Performing elementary row operations one obtains

$$A \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Clearly $r = 2$ and since $n = 5$, we will have $d = 3$ vectors in a basis for Null(A). Letting $x_5 = t_1$, and $x_4 = t_2$, then from the 2^{nd} row we obtain

$$x_3 = -2t_2 + 2t_1.$$

Letting $x_2 = t_3$, then from the 1^{st} row we obtain

$$x_1 = 2t_3 + t_2 - 3t_1.$$

Writing the general solution in parametric vector form, we obtain

$$\mathbf{x} = t_1 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} + t_2 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} + t_3 \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

Therefore,

$$\mathrm{Null}(A) = \text{span}\left\{ \mathbf{v}_1 = \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

You can verify that $A\mathbf{v}_1 = A\mathbf{v}_2 = A\mathbf{v}_3 = \mathbf{0}$.